

Size Generalisation of Neural Bellman-Ford: Aggregation and Step Supervision in a Small CPU Study

Rowan-67G

STEM seed corpus

<https://github.com/mihir-chauhan/research-harness-papers/tree/main/papers/neural-algorithmic-size-generalisation>

Abstract—We study how small networks trained to execute single-source Bellman-Ford shortest paths on random graphs with 4–8 nodes behave on graphs with 16, 32 and 64 nodes. We reimplement an MLP on the zero-padded flattened adjacency matrix and message-passing networks (MPNNs) with sum or max aggregation, each trained with or without supervision of the intermediate distance after every relaxation round, and test on two graph families: constant expected degree (sparse) and growing degree (dense). With five seeds and 1000 Adam steps per run, the expected pattern is only partly reproduced. On the registered primary metric, the mean error on sparse 64-node graphs, the sum-aggregation MPNN trained on the final distance alone is lowest and has no failing sparse seed. The max-aggregation MPNN trained on the final distance alone has the lowest error in distribution and the lowest median error at 64 nodes (medians were chosen post hoc), but two of its five seeds exceed our failure threshold there. The MLP is poor even in distribution, and its relative error at 64 nodes (0.943 on sparse graphs) is close to that of predicting zero. Step-wise supervision did *not* help in our setup: median error at 8 and 16 nodes was higher with it for both aggregators, and the sum-aggregation MPNN with step supervision produced a non-finite error on dense 64-node graphs in four of five seeds and an error of order 10^{17} in the fifth. None of the registered comparisons at 64 nodes is significant (all $p > 0.1$). For the two step-supervised MPNNs (three seeds each; the final-only models were not swept), running more message-passing steps at test time than $n - 1$ raised the error at 16 nodes in all six models and at 64 nodes in four of them, and running half as many steps lowered it. The study is small (one algorithm, one training budget, no tuning, five seeds) and the results should be read as a cautionary baseline, not a ranking.

Index Terms—neural algorithmic reasoning, size generalisation, message passing, Bellman-Ford, aggregation, intermediate supervision

I. INTRODUCTION

Neural algorithmic reasoning asks whether networks can learn to execute classical algorithms and keep working on inputs larger than those seen in training [1]. Graph algorithms are the standard testbed: the CLRS benchmark trains on small instances and tests on larger ones [2], and SALSA-CLRS pushes this to large sparse graphs [3]. Two design choices

are repeatedly credited for the success of message-passing networks on such tasks. The first is *algorithmic alignment*: the architecture should mirror the structure of the algorithm, which lowers sample complexity [4] and, for shortest paths, makes max (or min) aggregation a natural match for the relaxation step [5], [6]. The second is *step-wise supervision*: training the network to reproduce the algorithm’s intermediate states, as in neural execution of graph algorithms [7] and the hints of CLRS.

These two claims are usually evaluated inside larger systems that change several things at once. Here we isolate them for one algorithm, Bellman-Ford, in a 2×2 design (sum vs. max aggregation, with vs. without step supervision) plus an MLP that sees the flattened adjacency matrix. Networks are trained on graphs of at most 8 nodes and tested on 8, 16, 32 and 64 nodes, on two graph families that differ in whether node degree stays constant or grows with n ; degree shift is a known cause of size-generalisation failure in GNNs [8]. We state four falsifiable hypotheses (Section IV) before running and report all seeds.

Contributions. (i) A controlled, fully reproducible CPU-scale factorial study of aggregation and step supervision for size extrapolation of Bellman-Ford; (ii) a negative finding: step supervision did not help under our fixed budget, and sum aggregation with steps produced non-finite or astronomically large errors on growing-degree graphs at 64 nodes; (iii) a seed-level analysis showing that mean errors at 64 nodes are dominated by a minority of failing seeds, and a test-time step sweep of the two step-supervised systems (three seeds each) in which extra message-passing steps raise the error instead of leaving it unchanged. We claim nothing beyond this scale: one algorithm, small networks (the MPNN has 12,770 parameters and the MLP 557,376), five seeds, no hyperparameter search.

II. RELATED WORK

Neural execution and benchmarks. Veličković et al. [7] trained message-passing networks to imitate breadth-first search, Bellman-Ford and Prim step by step and argued for max aggregation; our max-aggregation, step-supervised system is a reimplementation of that recipe in spirit (our architecture and losses are our own simplified version, not their code). The CLRS benchmark [2] standardises 30 algorithms with hints

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and a 16-to-64 node size split, the triplet-GMPNN processor [9] and pointer graph networks [10] are other processors that we do not reimplement, and SALSA-CLRS [3] supplies sparse large-graph versions. We use a smaller 8-to-64 split on one algorithm.

Why aggregation should matter. Xu et al. [4] show that networks whose computation graph is close to the algorithm’s dynamic program need fewer samples, and Xu et al. [5] show that ReLU MLPs extrapolate linearly outside the training range and that GNNs with min/max aggregation can extrapolate on shortest-path-like tasks when the architecture matches. Dudzik and Veličković [6] relate GNNs to dynamic programming and discuss why max aggregation fits it. The message-passing framework itself is from Gilmer et al. [11], in the broader graph-network view of Battaglia et al. [12].

Size generalisation. Yehudai et al. [8] show that GNNs can fail to generalise to larger graphs because local neighbourhood structures (such as degree) differ between small and large graphs, which motivates our two graph families. Bevilacqua et al. [13] study size-invariant representations for graph classification, and Bevilacqua et al. later propose a causal self-supervised regulariser that improves out-of-distribution size generalisation on CLRS tasks [14]. We evaluate none of these remedies.

III. METHOD

Task. A graph has symmetric weighted adjacency $W \in \mathbb{R}^{n \times n}$ ($W_{ij} = 0$ if no edge, edge weights in $[0.2, 1]$), edge set E , and a source s . Bellman-Ford maintains $d_s^{(0)} = 0$, $d_i^{(0)} = \infty$ otherwise, and for $t = 1, \dots, n - 1$

$$d_i^{(t)} = \min \left(d_i^{(t-1)}, \min_{j:(i,j) \in E} (d_j^{(t-1)} + W_{ij}) \right). \quad (1)$$

All graphs are connected, so $d^{(n-1)}$ is finite. The network must output $d_i^{(n-1)}$ for all i .

MPNN (sum, mean or max). Node inputs are the source indicator $x_i = \mathbf{1}[i = s]$ (1 if i is the source, else 0); the weight enters only through edges. With $h_i^{(0)} = \text{enc}(x_i)$ and $T = n - 1$ steps,

$$m_{ij}^{(t)} = \phi([h_i^{(t-1)}, h_j^{(t-1)}, W_{ij}]), \quad (2)$$

$$a_i^{(t)} = \bigoplus_{j:(i,j) \in E} m_{ij}^{(t)}, \quad \bigoplus \in \{\sum, \text{mean}, \text{max}\}, \quad (3)$$

$$h_i^{(t)} = \psi([h_i^{(t-1)}, a_i^{(t)}]), \quad (\hat{d}_i^{(t)}, \ell_i^{(t)}) = \text{dec}(h_i^{(t)}), \quad (4)$$

where $\phi, \psi, \text{enc}, \text{dec}$ are two-hidden-layer ReLU MLPs of width 32 (hidden state 32), weights shared across steps. The max aggregator can implement Eq. (1) by letting ϕ output $-(d_j + W_{ij})$ coordinates, whereas sum aggregation must compute the minimum from a degree-dependent total. $\ell_i^{(t)}$ is a logit for “reached at round t ”.

Supervision. *Final-only:* $\mathcal{L} = \frac{1}{n} \sum_i (\hat{d}_i^{(T)} - d_i^{(T)})^2$. *Step-wise:* with $r_i^{(t)} = \mathbf{1}[d_i^{(t)} < \infty]$,

$$\mathcal{L} = \frac{1}{T} \sum_{t=1}^T \left[\frac{\sum_i r_i^{(t)} (\hat{d}_i^{(t)} - d_i^{(t)})^2}{\sum_i r_i^{(t)}} + \text{BCE}(\ell^{(t)}, r^{(t)}) \right], \quad (5)$$

so distances are supervised only where finite, and unreached nodes only through the reached flag.

MLP baseline. The weighted adjacency is zero-padded to 64×64 , flattened and concatenated with the padded one-hot source (4160 inputs), passed through two hidden layers of width 128 and a 64-way linear output (557,376 parameters, against 12,770 for each MPNN); the first n outputs are the predicted distances, trained with final-only MSE. Its input size forces the 64-node padding, so graphs with $n > 8$ put weight on inputs never active in training; this is the intended failure mode of a fixed-size model, not a tuned baseline. The MLP has no recurrent steps and therefore no step-supervised variant. All baselines are reimplemented here; none uses code from the cited papers.

IV. EXPERIMENTAL SETUP

Data. Each graph is a random spanning tree (each node attached to a uniformly random earlier node in a random order) plus independent Erdős–Rényi edges, undirected, weights $U(0.2, 1)$, uniformly random source. Training: $n \sim U\{4, \dots, 8\}$ per batch, ER probability $p = 0.4$, fresh graphs every step (no fixed training set). Test: 200 graphs per cell, $n \in \{8, 16, 32, 64\}$ and two families: *sparse* ($p = 3/(n - 1)$, constant expected extra degree) and *dense* ($p = 0.4$, degree grows with n). Cells are written S_n and D_n . The test graphs of seed k are drawn from a random stream seeded with $10000 + k$, which is separate from the training stream (seed k) and does not depend on the system. Each seed therefore has its own test set, and at a given seed every system, including the ablation and sweep runs, is evaluated on the same 200 graphs per cell; the paired tests pair on this seed.

Metrics. Primary: mean absolute error (MAE) of the predicted final distance over all nodes, cell S_{64} , averaged over seeds. Also relative error $\sum |\hat{d} - d| / \sum d$ (rS64, rD64; predicting zero everywhere gives exactly 1), and the fraction of nodes within 0.05. Because the distance scale differs between families (dense graphs have short paths), absolute errors are not comparable across families; the relative error is. A seed is counted as a *failure* in a cell if its MAE exceeds 1.0 or is non-finite (“div.” in the tables means a non-finite value, i.e. the network output overflowed). The threshold, the medians and the failure counts were introduced after seeing the first results; they are descriptive and not part of the registered tests.

Protocol. Five seeds (0–4) for all five main systems; three seeds (0–2) for ablations. Adam, learning rate 3×10^{-3} with cosine decay, 1000 steps, batch 32 graphs, gradient-norm clip 1, identical for every system. We did *no* hyperparameter tuning (the defaults were fixed before any run) and no test-size data was used for any choice; a validation MAE on the training distribution is logged but unused. All systems use the same data generators, metric code and number of message-passing steps ($T = n - 1$ at train and test, except in the step sweep of Section VI). Hardware: shared laptop CPU, 2 threads; the slowest run took about two minutes. The code, run log and per-seed metrics are in the repository.

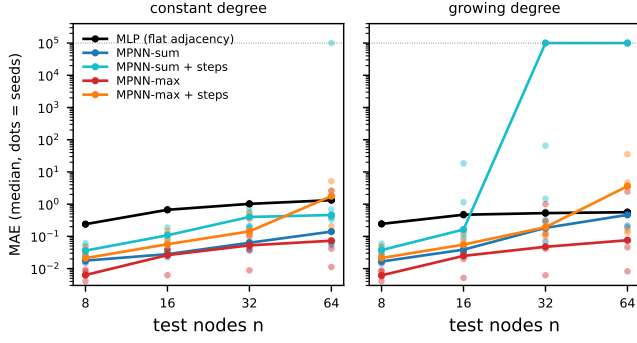


Fig. 1. MAE versus test size (log axes), median line and one dot per seed; left: sparse family, right: dense family. The axis is clipped: non-finite values and finite values of 10^5 or more (up to order 10^{17} , Table III) are all drawn at 10^5 (dotted line).

Hypotheses and registered tests (proposal.md, written before the runs). **H1** max+steps has lower S64 MAE than sum+steps; **H2** step supervision lowers the max-MPNN’s S64 MAE; **H3** the MLP is worse than every MPNN for $n \geq 16$ (tested at S16 and S64); **H4** under growing degree the per-seed ratio D64/D8 of sum+steps exceeds that of max+steps. H1–H3 are decided by the Welch and paired tests of `rh_compare` on seed-level MAE against the reference system max+steps, and count as not supported if the sign is wrong or the difference is not significant at 0.05. No test was registered for H4. Everything else we report is post hoc and marked as such: the S16 tests for H1 and H2, the final-only pair for H4, and the $\times 4$ column of the step sweep (the registered sweep was $\{0.5, 1, 2\}$).

V. RESULTS

Table I gives seed means $\pm$ standard deviations for the sparse family, Table II gives medians, relative errors and failure counts, and Table III lists every seed at 64 nodes, with the dense error at 8 nodes for the degree-shift comparison. Figure 1 shows all seeds at all sizes. On the registered primary metric, mean S64 MAE (Table I), the sum MPNN trained on the final distance is lowest (0.23 ± 0.19), followed by the max MPNN trained on the final distance (0.91 ± 1.22), the MLP, max+steps and sum+steps. These means are dominated by a minority of seeds (sum+steps seed 4 has S64 MAE of order 10^5 , Table III), so we also report medians and per-seed failure counts; that choice was made after seeing the data.

H1 (max+steps beats sum+steps at S64): not supported. The group means (Table IV) favour max+steps only because of sum+steps seed 4; the Welch and paired tests do not reject equality ($p = 0.374$), and the medians point the other way (Table II: S64 0.460 for sum+steps, 1.765 for max+steps, with 1/5 and 3/5 failing seeds). The post hoc S16 comparison is not significant either. On the growing-degree family the two systems fail differently (Table III): sum+steps gave a non-finite D64 error in 4/5 seeds and an error of order 10^{17} in the fifth, whereas max+steps exceeded the threshold in 3/5 seeds with

errors that stayed finite (at most 35.55). No test on D64 is possible with non-finite values.

H2 (step supervision helps max): not supported. In the registered S64 test the means are 0.909 without and 1.955 with steps ($p = 0.363$): the sign is opposite to H2 and the difference is within seed noise. The exploratory evidence points against H2: without steps the max MPNN has the lowest in-distribution error (S8 median 0.006 against 0.021 with steps) and a lower S16 error (paired $p = 0.041$ in a post hoc, uncorrected test on five seeds, Table IV). The medians at S8 and S16 are also lower without steps for sum (S8 0.018 vs. 0.036, S16 0.028 vs. 0.108). We do not know why: the step loss averages T terms plus a classification loss under the same step budget, so it may simply be optimised less on the final output; we did not run a loss-weight or longer-training control, so this is a statement about this budget only.

H3 (MLP worse than every MPNN for $n \geq 16$): supported at S16 against max+steps, not supported at S64. At S16 the MLP is far worse than the reference (MAE 0.671 vs. 0.078 for max+steps; Welch $p = 7.5 \times 10^{-7}$, paired $p = 4.2 \times 10^{-6}$); the other MPNNs were not tested against the MLP, but their S16 means lie between 0.03 and 0.12 (Table I). The MLP’s relative error at 64 nodes (rS64 0.943, rD64 0.926) is close to that of predicting zero, which is 1 by definition. At S64 its median 1.328 is *below* that of max+steps (1.765), and the test does not separate them ($p = 0.535$). The MLP also fits poorly in distribution (S8 0.239), so it gives no evidence about extrapolation per se; it may be under-trained at 1000 steps and we did not tune it.

H4 (sum degrades more than max under growing degree): direction as hypothesised, untested. Table III lists the two errors that form the per-seed ratio D64/D8; we do not print the quotient. For the registered pair, the step-supervised systems, the D8 errors are of similar size (medians 0.037 for sum+steps and 0.021 for max+steps), whereas the D64 error of sum+steps is non-finite in four seeds and of order 10^{17} in the fifth and that of max+steps stays finite (at most 35.55); the degradation from 8 to 64 nodes is therefore larger for sum in each of the five seeds. No test was registered and none is meaningful on non-finite values, so we report the direction and do not call H4 confirmed. For the final-only pair (post hoc) the per-seed values in Table III do not separate the two systems (their D64 errors overlap, with one non-finite seed for sum and two seeds above the threshold for max), and the failure counts at D64 point the other way (1/5 for sum, 2/5 for max). For that pair H4 is inconclusive.

Where it fails. Table III shows that every MPNN configuration has at least one failing seed in at least one family. The max MPNN without steps is accurate on three seeds (S64 MAE at most 0.073) but seeds 0 and 1 fail at both S64 and D64. The sum MPNN without steps has no failing sparse seed, but seed 4 gave a non-finite error at D64. Max+steps fails on seeds 1–3 in both families. Sum+steps fails on one sparse seed and on every dense seed. The non-finite errors come from non-finite network outputs; in the main group they occur only for sum aggregation and only on the dense family at 64 nodes, where

TABLE I
SPARSE FAMILY, MAE (MEAN $\pm$ STD OVER SEEDS; LAST COLUMN = NUMBER OF SEEDS); BOLD BEST, UNDERLINE SECOND.

Method	mae_sparse_n8 $\downarrow$	mae_sparse_n16 $\downarrow$	mae_sparse_n32 $\downarrow$	mae_sparse_n64 $\downarrow$	n
MPNN-sum + steps	0.04 $\pm$ 0.01	0.12 $\pm$ 0.04	0.43 $\pm$ 0.24	54707.72 $\pm$ 122329.15	5
MPNN-max	0.01 $\pm$ 0.00	0.03 $\pm$ 0.01	0.25 $\pm$ 0.46	<u>0.91 $\pm$ 1.22</u>	5
MPNN-sum	<u>0.02 $\pm$ 0.00</u>	<u>0.03 $\pm$ 0.01</u>	0.08 $\pm$ 0.06	0.23 $\pm$ 0.19	5
MLP (flat adjacency)	<u>0.24 $\pm$ 0.00</u>	<u>0.67 $\pm$ 0.01</u>	1.02 $\pm$ 0.02	1.33 $\pm$ 0.02	5
MPNN-max + steps	0.03 $\pm$ 0.01	0.08 $\pm$ 0.04	<u>0.24 $\pm$ 0.18</u>	1.96 $\pm$ 2.05	5

TABLE II
MEDIAN OVER 5 SEEDS. S/D: SPARSE/DENSE FAMILY, NUMBER = TEST NODES; MAE EXCEPT rS64/rD64 (RELATIVE ERROR). “fail” = SEEDS WITH MAE $>$ 1 OR NON-FINITE; DIV. = NON-FINITE.

System	S8	S16	S32	S64	D8	D64	rS64	rD64	fail S64	fail D64
MLP (flat adjacency)	0.239	0.667	1.020	1.328	0.244	0.562	0.943	0.926	5/5	0/5
MPNN-sum	0.018	0.028	0.063	0.140	0.017	0.464	0.097	0.763	0/5	1/5
MPNN-sum + steps	0.036	0.108	0.400	0.460	0.037	div.	0.324	div.	1/5	5/5
MPNN-max	0.006	0.026	0.053	0.073	0.006	0.076	0.051	0.127	2/5	2/5
MPNN-max + steps	0.021	0.057	0.143	1.765	0.021	3.611	1.223	6.055	3/5	3/5

TABLE III
PER-SEED VALUES (SEEDS 0–4): SPARSE MAE AT 64 NODES, DENSE MAE AT 64 NODES, AND DENSE MAE AT 8 NODES (H4 COMPARES THE LAST TWO). DIV. = NON-FINITE; VALUES OF 100 OR MORE ARE GIVEN TO ONE SIGNIFICANT FIGURE ($3e5 = 3 \times 10^5$).

System	S64 MAE, seed					D64 MAE, seed					D8 MAE, seed				
	0	1	2	3	4	0	1	2	3	4	0	1	2	3	4
MLP (flat adjacency)	1.32	1.33	1.32	1.36	1.33	0.57	0.55	0.59	0.54	0.56	0.25	0.25	0.24	0.24	0.24
MPNN-sum	0.42	0.45	0.063	0.14	0.054	0.46	0.19	0.58	0.22	div.	0.019	0.017	0.014	0.026	0.016
MPNN-sum + steps	0.33	0.46	0.39	0.67	3e5	div.	div.	div.	1e17	div.	0.050	0.037	0.023	0.032	0.059
MPNN-max	2.63	1.79	0.011	0.073	0.042	3.71	2.45	0.008	0.076	0.046	0.009	0.008	0.004	0.006	0.006
MPNN-max + steps	0.21	5.15	2.50	1.76	0.15	0.16	35.55	4.71	3.61	0.13	0.027	0.046	0.019	0.021	0.018

TABLE IV
TESTS AGAINST MPNN-MAX + STEPS (5 SEEDS, SEED-LEVEL MAE; “REF.” AND “OTHER” ARE THE GROUP MEANS OF MAX+STEPS AND OF THE COMPARED SYSTEM). ROWS MARKED * WERE NOT REGISTERED AND ARE POST HOC, UNCORRECTED.

	System	Cell	Ref.	Other	Welch p	paired p
H1	sum+steps	S64	1.955	5.5e+04	0.374	0.374
H1	sum+steps	S16*	0.078	0.119	0.160	0.231
H2	max	S64	1.955	0.909	0.363	0.363
H2	max	S16*	0.078	0.027	0.044	0.041
H3	MLP	S16	0.078	0.671	7.5e-07	4.2e-06
H3	MLP	S64	1.955	1.333	0.535	0.535

TABLE V
AGGREGATION ABLATION, MEDIAN MAE OVER SEEDS 0–2; IN PARENTHESES THE NUMBER OF SEEDS WITH MAE $>$ 1 OR NON-FINITE.

Aggregation	S8	S16	S64 (fail)	D64 (fail)
sum	0.016	0.026	0.417 (0/3)	0.464 (0/3)
mean	0.019	0.065	0.180 (0/3)	0.363 (1/3)
max	0.008	0.036	1.789 (2/3)	2.450 (2/3)
sum + steps	0.036	0.107	0.386 (0/3)	div. (3/3)
mean + steps	0.078	0.203	0.728 (0/3)	0.199 (0/3)
max + steps	0.026	0.081	2.502 (2/3)	4.709 (2/3)

a node sums over the most neighbours for the most steps. This may reflect activations growing with degree, but we logged no activations and no run isolates the cause. With five seeds we cannot rank the four MPNN configurations by reliability.

VI. ABLATIONS AND ANALYSIS

Aggregation type. Table V adds mean aggregation (3 seeds, medians; the sum and max rows use the same seeds 0–2 from the main group). Without steps, mean has the lowest S64 median (0.180) of the three, but with only three seeds, two of which fail for max, we do not claim a ranking. With steps,

mean has a higher S64 median than sum (0.728 against 0.386) and the highest S8 and S16 medians of all six rows, but it is the only step-supervised system with no failing seed at D64. One of the three mean-aggregation seeds without steps exceeds the failure threshold at D64. This ablation therefore shows that failures at 64 nodes are not confined to sum and max; it does not show that mean is better.

Test-time steps. Table VI and Fig. 2 rerun the two step-supervised systems (seeds 0–2) with the number of message-passing steps at test time scaled to 0.5, 1, 2 and 4 times $n - 1$ (training always used $n - 1$; the $\times 1$ column is the main-group result for the same seeds; $\times 4$ was added after the registered

TABLE VI

TEST-TIME STEP MULTIPLIER FOR THE STEP-SUPERVISED SYSTEMS: MAE PER SEED AND MEDIAN (SEEDS 0–2). VALUES OF 100 OR MORE TO ONE SIGNIFICANT FIGURE. †: NOT IN THE REGISTERED SWEEP (POST HOC).

System	Cell		$\times 0.5$	$\times 1$	$\times 2$	$\times 4^\dagger$
max+steps	S16	median	0.039	0.081	0.44	2.04
		seed 0	0.039	0.057	0.097	0.12
		seed 1	0.069	0.15	0.44	3.86
		seed 2	0.031	0.081	0.59	2.04
max+steps	S64	median	0.28	2.50	3e2	1e7
		seed 0	0.16	0.21	0.23	0.23
		seed 1	0.28	5.15	7e4	8e12
		seed 2	0.47	2.50	3e2	1e7
sum+steps	S16	median	0.067	0.11	0.19	0.35
		seed 0	0.077	0.11	0.19	0.35
		seed 1	0.067	0.12	0.61	0.80
		seed 2	0.054	0.073	0.11	0.21
sum+steps	S64	median	0.31	0.39	0.46	2.73
		seed 0	0.23	0.33	0.42	2.73
		seed 1	0.45	0.46	0.46	0.46
		seed 2	0.31	0.39	5.22	5e4

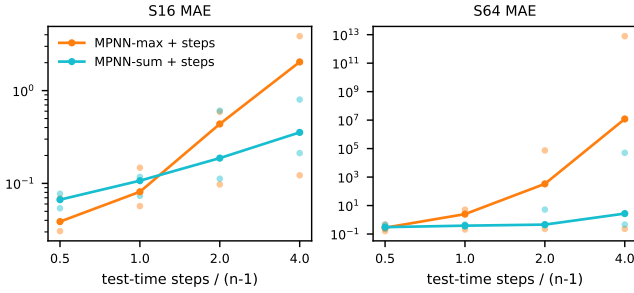


Fig. 2. Step-supervised systems: median MAE (dots: seeds 0–2) versus test-time steps divided by $n - 1$, sparse family, 16 and 64 nodes (the two panels have different vertical ranges). All values are finite and none is clipped.

sweep). The final-only systems were not swept, so nothing here applies to them. The median error increases with the multiplier in all four blocks: for max+steps at S64 it is 0.28 at $\times 0.5$ and 2.50 at $\times 1$, rising to 3×10^2 at $\times 2$ and 10^7 at $\times 4$; for sum+steps it rises more slowly (0.39 at $\times 1$, 2.73 at $\times 4$). Per seed the picture is less uniform. At S16 the error at $\times 2$ and at $\times 4$ is above that at $\times 1$ for all six models. At S64 four of the six models are clearly worse at $\times 4$ than at $\times 1$ (Table VI), but max+steps seed 0 (0.21 to 0.23) and sum+steps seed 1 (0.46 throughout) change little. Halving the steps lowers the error in every model and both cells. So for these six models extra rounds are not harmless: in most of them the learned update does not leave a finished computation unchanged, as the relaxation of Eq. (1) would. We did not measure the number of rounds Bellman-Ford needs to converge on these graphs (usually fewer than $n - 1$), so we cannot say whether $\times 0.5$ is better because it avoids accumulated error or because fewer rounds suffice; both are possible.

VII. LIMITATIONS

This is a small CPU study. (i) One algorithm, one graph generator and two test families; conclusions may not transfer to other CLRS algorithms, to integer weights, or to directed graphs. (ii) Five seeds for the main table and three for the ablations; the seed-to-seed spread is as large as most between-system differences, so most comparisons at 32 and 64 nodes are underpowered and only one registered test (MLP at S16) is significant. Medians, the failure threshold and the failure counts are descriptive choices made after seeing the data, and the post hoc tests are uncorrected. (iii) A single training budget of 1000 steps with no hyperparameter search for any system; the negative result for step supervision may reverse with longer training, different loss weights, teacher forcing or a different processor, and the MLP may be under-trained. (iv) The architectures are our simplified reimplementations (width 32, no gating, no layer norm, no triplet reasoning) of the cited ideas, not the published systems, so they say nothing about the published results. (v) Absolute errors are not comparable across the two families because distance scales differ; we ran no trivial reference predictor (such as a constant), so the only anchor for the relative error is that predicting zero gives 1. (vi) The training graphs are small enough that the training distribution is only partly representative (very few distinct graphs at $n = 4$). (vii) The step sweep covers only the two step-supervised systems on three seeds; whether the final-only models tolerate extra steps is unknown. (viii) The mechanism behind the failures (accumulating error, activation growth with degree, non-idempotent updates) is not isolated by any run in this paper, and the runs with non-finite outputs were not inspected beyond their final metrics. A full study would add more seeds, tuned baselines, larger training sizes, other algorithms and the published processors.

VIII. CONCLUSION

In this small study of neural Bellman-Ford with training on at most 8 nodes, no registered comparison at 64 nodes was statistically significant. On the registered primary metric (mean error on sparse 64-node graphs) the sum MPNN trained on the final distance alone was lowest and had no failing sparse seed, although one of its seeds gave a non-finite error on dense graphs. The aggregation-aligned max MPNN trained on the final distance alone had the lowest error in distribution and, by the post hoc median, at 64 nodes, but it failed on two of five seeds there. Step-wise supervision did not help under our fixed budget, and the sum MPNN with it gave non-finite or astronomically large errors on growing-degree graphs at 64 nodes. The flattened-adjacency MLP was poor at every size, including in distribution. For the two step-supervised systems (three seeds each), running more message-passing steps than $n - 1$ at test time raised the median error, and the error of most individual models, so these processors mostly do not behave as stable fixed-point iterations; the final-only models were not swept. The intuition that alignment plus hints yields size generalisation was therefore not confirmed here; a larger

and tuned study is needed before drawing a conclusion either way.

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